

Sound And Its Speed

ACOUSTICS:

The study of Production and properties of sound is called Acoustics. It also deals with possible sources of distortions and their effect on the quality of sound is called acoustics.

SOUND:**Definition**

A vibration that propagates as an audible wave of pressure, through a medium such as a gas, liquid or solid. It is produced by vibrating bodies. It travels in the form of longitudinal waves and require a medium.

Sound can be classified as

(a) **NOISE**

A sound which produces unpleasant effect on the ear is known as noise. It is a sound of irregular frequency. It is provided by a body producing irregular vibrations.
e.g.: Slamming of door, rumbling of a truck etc.

(b) **MUSICAL SOUND**

A sound which produces pleasant effect on the ear is known as musical sound. It is a sound of regular frequency. It is caused by a body producing regular vibrations.
e.g.: Sound produced by piano or guitar

SPEED OF SOUND:

Sound travels in form of compression and rarefaction. The speed of sound waves depends upon

- (i) Compressibility of medium
- (ii) Inertia property of medium

Newton proposes the following formula for speed of sound waves

$$v = \sqrt{\frac{\text{Compressibility of medium}}{\text{Inertial property of medium}}}$$

Speed of Sound in Solids:

$$v = \sqrt{\frac{E}{\rho}}$$

Speed of Sound in Liquids:

$$v = \sqrt{\frac{\beta}{\rho}}$$

Where

ρ = Density of medium.

β (Bulk Modules) is defined as "The ratio in change in pressure to volumetric strain"

Newton assumed that the increase in temperature in compression area is promptly neutralized by the decrease in the rarefaction zone and for all practical purposes the temperature of the medium (air) remains constant. This means sound travel through air under isothermal conditions and under such conditions Boyle's law is applicable.

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If the initial pressure is P and ΔP is the small increase in pressure so that the final pressure is $P + \Delta P$. This increment in pressure causes a decrease in the volume, say by ΔV , then the volume changes (decreases) from the initial value V to final value $V - \Delta V$. Boyle's law states:

$$\begin{aligned} PV &= (P + \Delta P)(V - \Delta V) \\ PV &= PV - P\Delta V + V\Delta P - \Delta P\Delta V \\ PV - PV &= -P\Delta V + V\Delta P - \Delta P\Delta V \\ 0 &= -P\Delta V + V\Delta P - \Delta P\Delta V \\ P\Delta V &= V\Delta P - \Delta P\Delta V \end{aligned}$$

If ΔP is very small, so is ΔV and $\Delta P\Delta V$ is even smaller. Therefore this term can be neglected. The above equation becomes;

$$P\Delta V = V\Delta P$$

$$P = \frac{V\Delta P}{\Delta V}$$

$$P = \Delta P \times \frac{V}{\Delta V}$$

$$P = \Delta P \div \frac{\Delta V}{V}$$

$$P = \frac{\Delta P}{\frac{\Delta V}{V}}$$

However, ΔP is stress and $\frac{\Delta V}{V}$ is volumetric strain. Therefore,

$$P = \frac{\Delta P}{\frac{\Delta V}{V}} = \frac{\text{stress}}{\text{volumetric strain}} = \beta$$

β is known as modulus of elasticity for isothermal process. Put this value of β in equation (i)

$$v = \sqrt{\frac{P}{\rho}} \quad \text{--- (ii)}$$

This is the equation of sound wave in air derived by Newton.

Calculation based on Newton's assumption:

Using Mercury Barometer, the pressure P can be calculated as

$$P = \frac{F}{A} = \frac{mg}{A} = \frac{\rho_m Ahg}{A} = \rho_m gh$$

Use this value of P in equation (ii),

$$v = \sqrt{\frac{\rho_m gh}{\rho}}$$

Where;

$$\rho = \frac{m}{V}$$

$$m = \rho V$$

$$\text{As } V = Ah$$

$$m = \rho Ah$$

We know that

$$\rho_m = 13.6 \text{ g/cm}^3$$

$$g = 980 \text{ cm/sec}$$

$$h = 76 \text{ cm}$$

$$\rho = \text{Density of air} = 0.001293 \text{ gm/cm}^3$$

Using these values,

$$v = \sqrt{\frac{13.6 \times 980 \times 76}{0.001293}} = 28100 \frac{\text{cm}}{\text{s}} = 281 \text{ m/s}$$

However, experimentally proved speed of sound is 332 m/s. Thus the theoretical value (281 m/s) is 16% less than the experimental value (332 m/s). This difference is big enough to be accepted!

Error in Newton's Formula

Newton assumed that sound waves propagate through fluids under isothermal conditions i.e. compressions and rarefactions occur at the same temperature, which is not true. Laplace corrected this concept by proposing that the propagation of waves is an adiabatic process rather than an isothermal process.

Laplace Correction:

In 1816 pointed out that due to

- Sudden nature of compressions and rarefaction of sound waves
- Bad conducting nature of air/gas.

Temperature of gas is not constant throughout. It rises at compression and falls at rarefaction. Thus compression and rarefaction occur adiabatically; there is no flow of heat into or out of gas. Its total heat content remains same.

Applying Boyle's Law:

$$PV^\gamma = \text{Constant}$$

Here γ is a constant depending on the nature of gas.

$$PV^\gamma = (P + \Delta P)(V - \Delta V)^\gamma \Rightarrow PV^\gamma = (P + \Delta P) \left[V \left(1 - \frac{\Delta V}{V} \right) \right]^\gamma = (P + \Delta P) \cdot V^\gamma \left(1 - \frac{\Delta V}{V} \right)^\gamma$$

Cancelling the common factor on both sides,

$$P = (P + \Delta P) \left(1 - \frac{\Delta V}{V} \right)^\gamma$$

Applying binomial theorem and neglecting very small terms;

$$(1 + x)^n = 1 + \frac{nx}{1!} + \frac{n(n-1)x^2}{2!} + \dots \text{ (Binomial Theorem)}$$

$$P = (P + \Delta P) \left(1 - \frac{\gamma \Delta V}{V} \right) = P - \gamma P \frac{\Delta V}{V} + \Delta P$$

(Smaller terms are neglected).

Cancelling P on both sides and re-arranging,

$$\Delta P = \gamma P \frac{\Delta V}{V} \quad \text{OR} \quad \gamma P = \Delta P \div \frac{\Delta V}{V} = \frac{\Delta P}{\frac{\Delta V}{V}}$$

The R.H.S is Adiabatic Bulk Modulus, β . Therefore,

$$\gamma P = \beta$$

Therefore, substituting for β in equation (i),

$$v = \sqrt{\frac{\beta}{\rho}} = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma \rho_m g h}{\rho}}$$

Putting the values,

$$v = \sqrt{\frac{1.42 \times 13.6 \times 980 \times 76}{0.001293}} = \sqrt{\frac{1438357.76}{0.001293}}$$

$$= \sqrt{1112418994.586234} = 33353 \frac{\text{cm}}{\text{s}} \Rightarrow v = 333.53 \text{ m/s}$$

This value is in agreement with the experimentally calculated value of 332 m/s.

Factors Effecting Speed of Sound:

Effects of Pressure

So the Laplace Formula for the speed of sound in air is

$$v = \sqrt{\frac{\gamma P}{\rho}}$$

As Pressure Increase/decreases, density also increases/decreases in same proportion i.e., ratio P/ρ remain constant so the speed of sound is not affected by changing pressure.

Effect of Temperature

As we know that

$$\rho = \frac{\text{Mass}}{\text{Volume}} = \frac{m}{V}$$

Substitute the value of ρ in Laplace formula of speed of sound,

$$v = \sqrt{\frac{\gamma n R T / V}{m / V}}$$

$$v = \sqrt{\frac{\gamma R T}{m / n}}$$

Where, $m/n = M$ called Molar mass of the gas (air).

$$v = \sqrt{\frac{\gamma R T}{M}}$$

Speed of sound is directly proportional to the square root of absolute temperature of gas

$$v \propto \sqrt{T}$$

Effect of Density Or Nature Of Medium

At constant pressure velocity of sound varies inversely as the square root of density of the medium. Thus, speed of sound is fastest in solids, less in liquids and least in air.

$$v = \sqrt{\frac{\gamma P}{\rho}}$$

$$v = \sqrt{\frac{\text{Constant}}{\rho}} \quad [\because \text{At Constant Pressure}]$$

$$v \propto \frac{1}{\sqrt{\rho}}$$

Examples

Velocity of sound in H_2 is 4 times the velocity in O_2 , as oxygen is 16 times denser than hydrogen.

Effect Of Wind And Moisture:

- The presence of moisture in the air decreases the resultant density of air which increases the speed of sound in humidity. Hence the speed of sound in damp (wet) air is greater than in dry air.
- If v_w is the wind speed then the speed of sound along the direction of wind relative to ground is $(v + v_w)$ and $(v - v_w)$ against the direction of wind

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Superposition Of Waves

SUPERPOSITION:

The Arrival of two or more waves traveling in a medium at a common point is called superposition of waves.

INTERFERENCE:

The resultant effect of superposition of two waves is called interference. There are two types of interference.

1. Constructive interference
2. Destructive interference

Principle of Superposition:

When two or more wave traveling nearly linearly in a medium arrive at a point simultaneously the net displacement of the medium is equal to the algebraic sum of displacement of all the waves.

$$Y = Y_1 + Y_2$$

Analytical Treatment

Consider two harmonic waves of same frequency wavelength and amplitude .but of different phase are traveling in same direction in a medium.

$$Y_1 = A_0 \sin (Kx - \omega t)$$

$$Y_2 = A_0 \sin (Kx - \omega t - \phi)$$

$$Y = Y_1 + Y_2$$

$$Y = A_0 \sin (Kx - \omega t) + A_0 \sin (Kx - \omega t - \phi) \quad \text{---(1)}$$

Or

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Now according to the trigonometric formulas

$$\sin \alpha + \sin \beta = 2 \cos \left(\frac{\alpha - \beta}{2} \right) \sin \left(\frac{\alpha + \beta}{2} \right)$$

Here $(Kx - \omega t) = \alpha$, $(Kx - \omega t - \phi) = \beta$

So by using above formula in equation (1)

$$Y = 2 A_0 \cos \left[\frac{(kx - \omega t) - (kx - \omega t - \phi)}{2} \right] \sin \left[\frac{(kx - \omega t) + (kx - \omega t - \phi)}{2} \right]$$

$$Y = 2 A_0 \cos \left[\frac{\phi}{2} \right] \sin \left[\frac{(2kx - 2\omega t - \phi)}{2} \right]$$

This equation shows that Resultant wave is also harmonic and has same frequency and wavelength, as individual waves. Amplitude of resultant wave only depends upon $2 A_0 \cos (\phi/2)$

Phase of resultant wave is $\phi/2$.

- For maximum amplitude

$$\cos (\phi/2) = 1$$

or $\phi = 0$

Then amplitude

$$2 A_0 (1) = 2 A_0$$

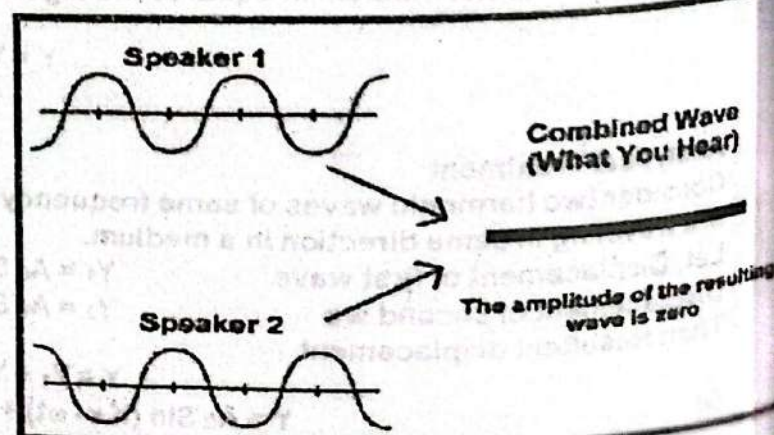
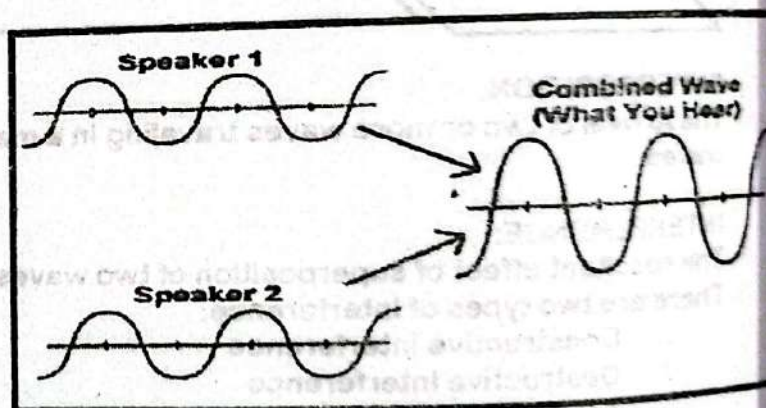
This means the amplitude of resultant wave is twice as large as that of individual waves. This happens when crests or troughs of two waves fall on each other. This is called Constructive Interference. It takes place at

$\phi = 0, 2\pi, 4\pi, \dots$ etc.

- For minimum amplitude

$$\cos (\phi/2) = 0 \quad \text{or} \quad \phi = \pi$$

Then amplitude of resultant wave is zero. This happens when crests of one wave fall on troughs of other wave. This is called Destructive Interference. Destructive interference takes place when $\phi = \pi, 3\pi, 5\pi, \dots$ etc.



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Beats

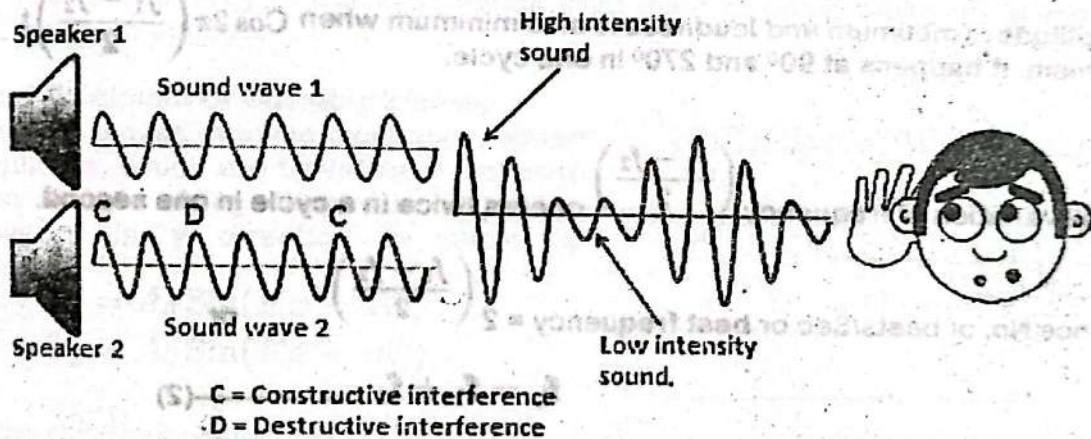
BEATS:

Periodic rise and fall in the loudness or intensity of sound is called Beat,
OR

When two sound waves of slightly different frequencies approach your ear, you hear a fluctuating sound (soft and loud sound alternatively). This phenomenon is known as beats. The loud sound is caused by the constructive interference and the soft sound is caused by the destructive interference.

EXPLANATION

When two sound waves of same amplitude and slightly different frequencies reach in human ear, human ear does not hear two different sound, but hear a single sound which rises and fall in intensity periodically. This phenomenon is called beats. One rise and one fall and then again rise of sound is known as one beat. The number of beats occurring per second is called *beat frequency* (f_b). The time interval between successive loud sound or low sound is called *beat period*.

**Analytical Treatment**

Consider two sound wave of slightly different frequencies f_1 and f_2 but of same amplitude

Then $\omega_1 = 2\pi f_1$ & $\omega_2 = 2\pi f_2$

The displacement of two waves at any instant will be

$$Y_1 = A_0 \cos \omega_1 t \quad \text{or}$$

$$Y_1 = A_0 \cos 2\pi f_1 t$$

and

$$Y_2 = A_0 \cos \omega_2 t \quad \text{or}$$

$$Y_2 = A_0 \cos 2\pi f_2 t$$

When two such wave superpose with each other the displacement of resultant wave will be

$$Y = Y_1 + Y_2$$

$$Y = A_0 \cos(2\pi f_1 t) + \cos(2\pi f_2 t)$$

$$Y = A_0 [\cos(2\pi f_1 t) + (\cos 2\pi f_2 t)] \quad \text{--- (1)}$$

But by trigonometry

$$\cos \alpha + \cos \beta = 2 \cos \left(\frac{\alpha - \beta}{2} \right) \cos \left(\frac{\alpha + \beta}{2} \right)$$

Applying on equation (1)

$$Y = A_0 \left[2 \cos \left(\frac{2\pi f_1 t - 2\pi f_2 t}{2} \right) \cos \left(\frac{2\pi f_1 t + 2\pi f_2 t}{2} \right) \right]$$

This equation shows that

$$Y = 2 A_0 \left[\cos 2\pi \left(\frac{f_1 - f_2}{2} \right) t \cos 2\pi \left(\frac{f_1 + f_2}{2} \right) t \right]$$

(1) Resultant wave is also harmonic with frequency $\frac{f_1 + f_2}{2}$ It is called effective or average frequency.

(2) Amplitude of resultant wave is $A = 2A_0 \cos 2\pi \left(\frac{f_1 - f_2}{2} \right) t$

This equation show that amplitude of resultant sound wave is not constant it varies with time (t)

Conclusion

(1) Amplitude is maximum and loudness of resultant sound is maximum when

$\cos 2\pi \left(\frac{f_1 - f_2}{2} \right) t$ is maximum. It happens at 0° and 360° in one cycle.

(2) Amplitude is minimum and loudness is also minimum when $\cos 2\pi \left(\frac{f_1 - f_2}{2} \right) t$ minimum. It happens at 90° and 270° in one cycle.

(3) Thus variation of frequency $\left(\frac{f_1 - f_2}{2} \right)$ occurs twice in a cycle in one second.

Hence No. of beats/Sec or beat frequency = $2 \left(\frac{f_1 - f_2}{2} \right)$ or

$$f_b = f_1 \pm f_2 \text{ -----(2)}$$

Equation (2) shows that numbers of beats heard per second is equal to the difference in frequency of two sound waves.

Human ear can hear 7 beats/sec .so the beat frequency for human ear is 7 beats/sec

Precaution:

The difference in frequencies of the two sounds should not exceed 16; otherwise the beats are so continuous that one cannot distinguish between them.

Applications

- 1- In tuning a musical instrument.
- 2- In measuring unknown frequency of a tuning fork.

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Stationary Waves

STATIONARY WAVES :

When two waves of equal amplitude and frequency traveling through the same medium in opposite direction, superimpose, the result in a wave which does not travel in either direction. Such a wave is called Standing wave.

Characteristics

1. A medium is necessary.
2. They are transverse in nature.
3. They appear to be stand still.
4. Energy is not transferred through the medium by them.
5. There have fixed positions of zero amplitude called **NODES**.
6. The points of maximum amplitude are called **ANTINODES**.
7. The enclosed area b/w two nodes is called loop.
8. Length of two loops is one Wavelength (λ).
9. The total energy of particles varies from point to point, being maximum at antinodes and minimum at node.

Analytical Treatment Of Standing Waves:

Consider two waves of same frequency, speed and amplitude, which are traveling in opposite direction along a string. Displacement of two such waves in y direction is given by equation's

$$Y_1 = A_0 \sin(Kx - \omega t)$$

$$Y_2 = A_0 \sin(Kx + \omega t)$$

Hence the resultant wave after super position may be written as:

$$Y = Y_1 + Y_2$$

$$Y = A_0 \sin(Kx - \omega t) + A_0 \sin(Kx + \omega t)$$

$$Y = A_0 [\sin(Kx - \omega t) + (\sin Kx + \omega t)]$$

Since we know that, according to trigonometry,

$$\sin(\alpha + \beta) + \sin(\alpha - \beta) = 2\sin \alpha \cos \beta$$

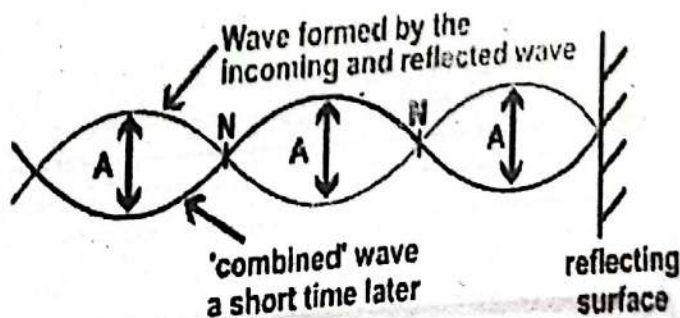
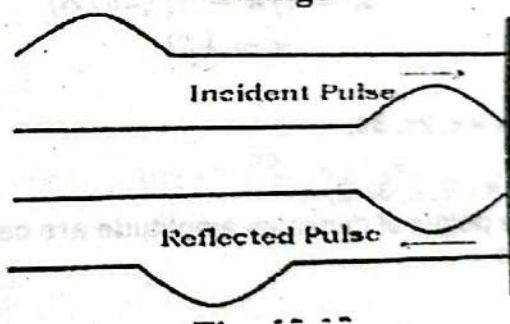
$$\alpha = Kx$$

$$\beta = \omega t$$

$$Y = A_0 [2 \sin Kx \cdot \cos \omega t]$$

$$Y = 2A_0 \sin Kx \cdot \cos \omega t$$

Reflection of Waves and Phase change



Conclusions

This equation shows that

- (1) Standing waves have constant angular velocity (ω).
- (2) The amplitude of particles of stationary wave varies and is given by $[2 A_0 \sin Kx]$ i.e depends on $[\sin Kx]$.

Antinodes:

Amplitude will be maximum = $2 A_0$, when

$$\sin Kx = 1 \quad \text{i.e}$$

$$Kx = \sin^{-1}(1)$$

$$\text{As } Kx = \pi/2$$

$$\therefore x = \pi/2k = \pi/2(2\pi/\lambda)$$

$$x = \lambda/4$$

$$Kx = \pi/2, 3\pi/2, 5\pi/2; \text{---}$$

or

$$x = \lambda/4, 3\lambda/4, 5\lambda/4; \text{---}$$

These points of maximum amplitude are called Antinodes.

Nodes:

Amplitude will be minimum = 0 When

$$\sin Kx = 0 \quad \text{i.e}$$

$$Kx = \sin^{-1}(0)$$

$$\text{As } Kx = \pi$$

$$\therefore x = \pi/k = \pi/(2\pi/\lambda)$$

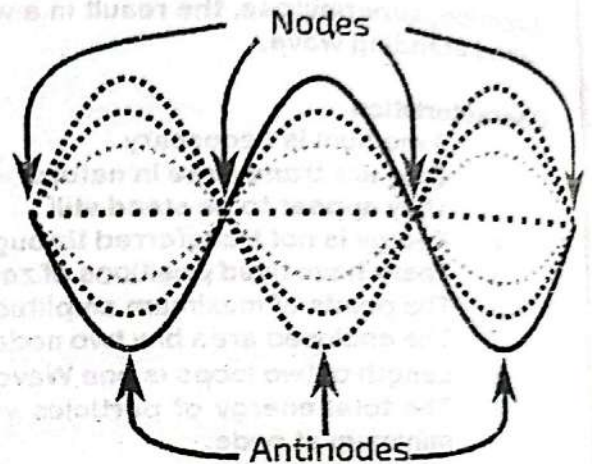
$$x = \lambda/2$$

$$Kx = \pi, 2\pi, 3\pi; \text{---}$$

or

$$x = \lambda/2, \lambda, 3\lambda/2; \text{---}$$

These points of minimum amplitude are called Nodes



Separation of two adjacent nodes or two adjacent antinodes is $\lambda/2$

Separation b/w two consecutive node and antinodes is $\lambda/4$.

Examples Of Standing Waves

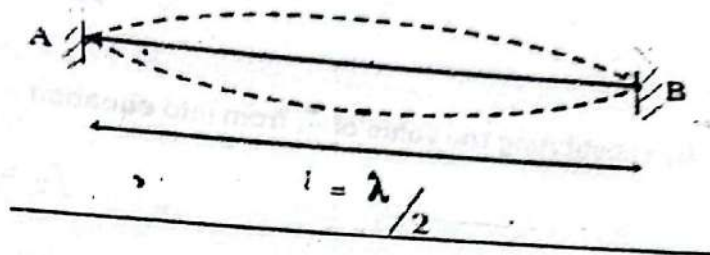
1. Waves on stringed instruments.
2. Waves inside organ pipes, or wind instruments and resonance tube.
3. Light waves inside a LASER.

Modes Of Vibration Of Stretched String

MODES OF VIBRATION IS A STRECHED STRING:

FUNDAMENTAL MODE or FIRST HARMONIC:

If the string is plucked at its middle point and then released, two transverse waves are set up in the string moving in opposite directions. At the ends reflection takes place and stationary waves are produced. The string vibrates in one loop (Figure). If f_1 and λ_1 be the frequency and wavelength of either of the wave respectively, then from figure we see that:



$$L = \frac{\lambda_1}{2}$$

$$\lambda_1 = 2L$$

If v is the speed of either of the component waves, then:

$$v = f_1 \lambda_1$$

$$v = f_1 2L$$

$$f_1 = \frac{v}{2L}$$

This is called fundamental frequency or 1st harmonic.

If m is the total mass of the string, then it can be shown that the velocity v of the wave along the string is given by:

$$v = \sqrt{\frac{T}{\mu}}$$

Where μ = Linear density of the string

Putting the value of v in equation

$$f_1 = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

Where,

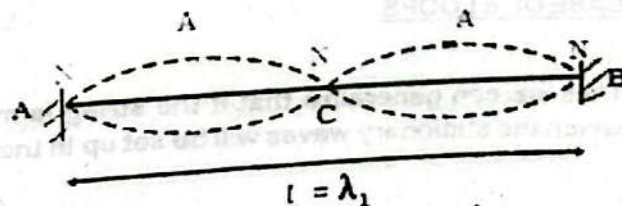
$$\mu = \frac{m}{l}$$

SECOND HARMONIC or FIRST OVERTONE:

If f_2 and λ_2 be the frequency and Wavelength of one of the component.

Waves, then from figure we can see that

$$\lambda_2 = L$$



The speed will be same since it does not

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depend on the number of loops. Therefore:

$$v = f_2 \lambda_2$$

$$v = f_2 L$$

$$f_2 = \frac{v}{L}$$

$$f_2 = 2\left(\frac{v}{2L}\right)$$

By substituting the value of f_1 from into equation .

$$f_2 = 2f_1$$

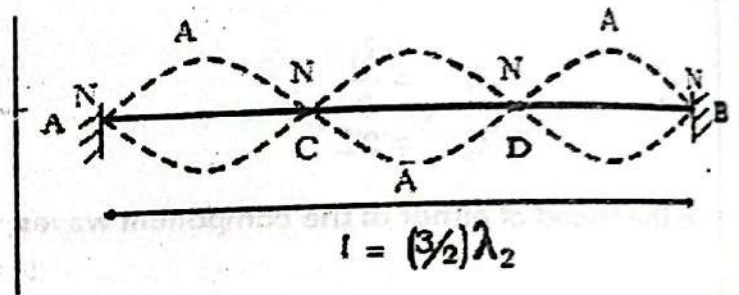
This is called 2nd harmonic and first overtone

THIRD HARMONIC or SECOND OVERTONE:

In this case string will vibrate in three loops.
From fig , we can see that:
also

$$\lambda_3 = \frac{2L}{3}$$

$$v = f_3 \lambda_3$$



Putting the value of λ_3

$$v = f_3 \times \frac{2L}{3}$$

$$f_3 = \frac{3v}{2L}$$

Comparing equation, we have:

$$f_3 = 3f_1$$

This is called 3rd harmonic and 2nd overtone

CASE OF n LOOPS

Thus we can generalize that if the string is made to vibrate in n loops, and then the frequency at which the stationary waves will be set up in the string will be:

$$f_n = n f_1$$

The lowest of these frequencies i.e. f_1 is known as the fundamental and the others which are the integral multiples of the fundamental are known as overtones or harmonies.

FUNDAMENTAL FREQUENCY AND OVERTONES

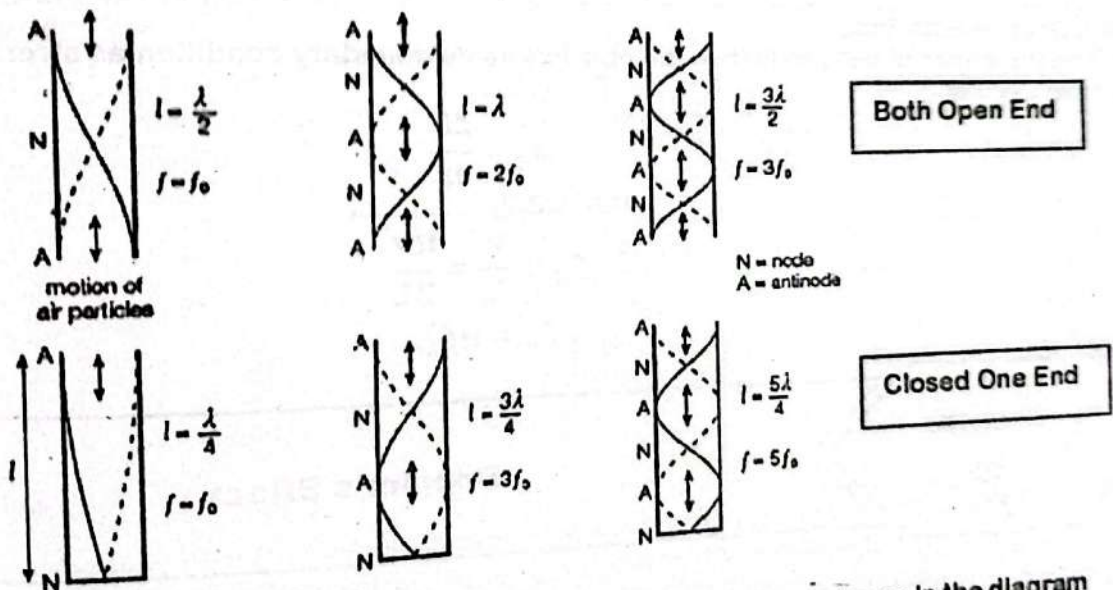
- Lowest frequency of stationary waves is called FUNDAMENTAL.
- Frequency which is integral multiple of fundamental are called OVERTONES

STATIONARY SOUND WAVES:

Sound waves are three-dimensional waves that propagate in all directions. However, when sound waves are confined in a pipe, they can form standing waves. This is because the waves are reflected at the boundaries of the pipe, and the reflections interfere with the incoming waves to create a standing wave pattern.

The air column inside a pipe can be open at both ends or at one end only. If the pipe is open at both ends, the standing waves will have antinodes (points of maximum displacement) at the open ends and nodes (points of zero displacement) in the middle of the pipe. If the pipe is open at one end only, the standing waves will have an antinode at the open end and a node at the closed end. The frequency of the standing waves in a pipe is determined by the length of the pipe and the speed of sound in air. The shorter the pipe, the higher the frequency of the standing waves.

Organ Pipes and flutes are musical instruments that use standing waves to produce sound. The length of the pipes in these instruments is carefully chosen to produce the desired frequencies of sound.



Pipes Closed At One End

If the organ pipe is closed at one end than closed end will be an antinode as shown in the diagram and we can say that:

First Harmonic:

$$L = \frac{\lambda}{4} \text{ or } \lambda = 4L$$

Which is twice as large as the wavelength ($2L$) of fundamental mode of vibration in a pipe of same length, open at both ends.

Since;

$$f_1 = \frac{v}{\lambda_1} = \frac{v}{4L}$$

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Second Harmonic:

If the air column vibrates for second harmonic or first overtone, then length of pipe can be given as

$$L = \frac{\lambda_2}{2} + \frac{\lambda_2}{4} = \frac{3\lambda_2}{4}$$

Since; $\lambda_2 = \frac{4L}{3}$

$$f_2 = \frac{3v}{4L}$$

Or, $f_2 = 3f_1$

Similarly for third harmonic or second overtone

$$L = \frac{5\lambda_3}{4}$$

And $\lambda_3 = \frac{4L}{5}$ and $f_3 = 5f_1$

Note that the standing wave frequencies for closed pipe at one end are odd integral multiple of the fundamental frequency.

$$f_n = \frac{v}{\lambda_n} = \frac{nv}{4L} \quad (\text{as } \lambda_n = 4L/n)$$

Generally:

$$f_n = (2n - 1) f_1$$

Where $n = 1, 2, 3, 4, 5, \dots$ **Pipe Open At Both Ends**

If the pipe is open at both ends then the pipe has same boundary condition as stretched string and we can say that:

$$\lambda_n = \frac{2L}{n}$$

$$n = 1, 2, 3, \dots$$

$$f_n = \frac{v}{\lambda_n} = \frac{nv}{2L}$$

$$f_n = nf_1$$

Chapter 12

Doppler's Effect

DOPPLER'S EFFECT:

The Doppler effect (or Doppler shift), named after Austrian physicist Christian Doppler who proposed it in 1842,

Definition

The apparent change in pitch or frequency of sound due to relative motion of either source or listener or both is called DOPPLER'S EFFECT.

Examples

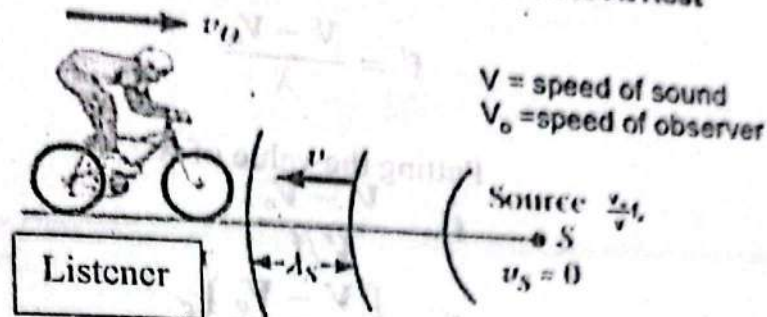
A person standing on a platform hears the increasing pitch of whistle of engine as it approaches him and when the engine moves away the pitch decreases. Same happens when source is stationary and listener moves to it or away from it.

Reason

This is because the apparent frequency (f') of sound waves received by listener is either greater Or lesser than the actual frequency (f) depending upon the relative motion of source or listener.

CASE 1: When The Listener Is Moving And The Source Is At Rest:

(A) When The Listener Is Moving Towards The Source Of Sound Is At Rest



Suppose the listener is moving towards a stationary source of sound. The speed of the listener is V_o . Let the source emits sound waves of frequency (f), speed (V) and wave length equal to λ . We know that

$$\lambda = V/f \quad (1)$$

The sound waves emitted by the source spread out in all directions, with each wave crest separated by the wavelength λ . When the listener moves towards the source, the speed of the sound waves relative to the listener increases to $(V + V_o)$. As a result, the listener perceives a higher frequency, denoted as f' .

$$f' = \frac{V + V_o}{\lambda}$$

Putting the value of λ

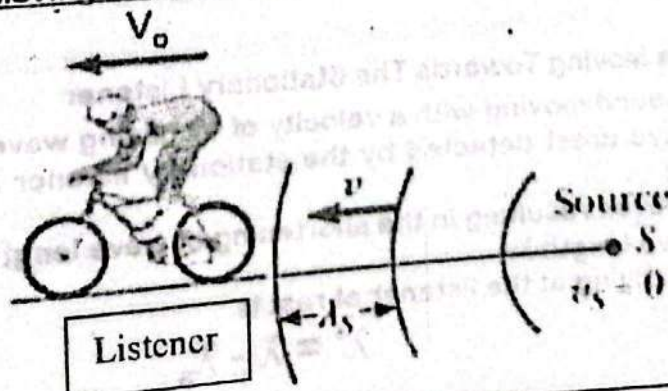
$$f' = \frac{V + V_o}{V/f}$$

$$f' = \left\{ \frac{V + V_o}{V} \right\} f$$

Conclusion

This expression shows that when a listener moves towards a stationary source of sound, he detects larger frequency and hence higher pitch as compared to original.

(B) When The Listener Is Moving Away From The Stationary Source



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Suppose the listener is moving away from a stationary source of sound. The speed of the listener is V_o . Let the source emits sound waves of frequency f , speed (V) and wave length equal to λ . We know that

$$\lambda = V/f \quad (1)$$

The sound waves emitted by the source spread out in all directions, with each wave crest separated by the wavelength λ . When the listener moves away from the source, the speed of the sound waves relative to the listener decreases to $(V - V_o)$. As a result, the listener perceives a lower frequency, denoted as f' .

$$f' = \frac{V - V_o}{\lambda}$$

Putting the value of λ

$$f' = \frac{V - V_o}{V/f}$$

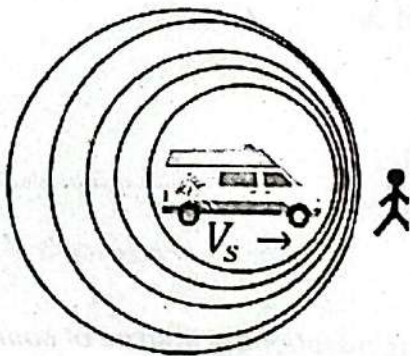
$$f' = \left\{ \frac{V - V_o}{V} \right\} f$$

Conclusion:

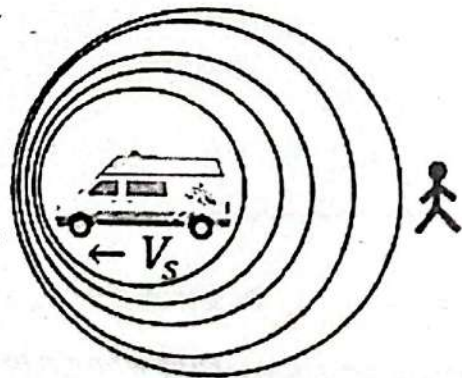
This expression shows that when a listener moves away from a stationary source of sound, he detects lower frequency and hence low pitch as compared to original.

CASE 2 : When The Source Is Moving And The Listener Is At Rest.

Source is moving towards
The observer who is at rest



Source is moving away from
The observer who is at rest

**(A) When The Source Is Moving Towards The Stationary Listener**

Consider a source of sound moving with a velocity of V_s having wavelength (λ_s) and sounding a tone. source is moving in the direction of outgoing waves resulting in the shortening of wave length i.e. λ' measured by the listener is shorter than true wave length λ .

Thus the wave length arriving at the listener at rest is

$$\lambda' = \lambda - \lambda_s$$

$$\begin{aligned}\frac{V}{f'} &= \frac{V}{f} - \frac{V_s}{f} \\ \frac{V}{f'} &= \frac{V - V_s}{f} \\ \frac{f'}{V} &= \frac{f}{V - V_s} \\ f' &= \left\{ \frac{V}{V - V_s} \right\} f\end{aligned}$$

Conclusion

This expression indicates that the frequency of sound as heard by the listener is increased and he will detect a higher pitch of sound.

(B) When the Source is Moving Away From the Stationary Listener

Consider a source of sound moving with a velocity of V_s and sounding a tone of frequency f . The wave crest detected by the stationary listener are wide spread because the source is moving away from the direction of incoming waves resulting in the increase of wave length i.e. λ' measured by the listener is longer than true wave length λ .

Thus the wave length arriving at the listener at rest is

$$\lambda' = \lambda + \lambda_s$$

Putting the value of λ i.e. $\lambda = V/f$

$$\begin{aligned}\frac{V}{f'} &= \frac{V}{f} + \frac{V_s}{f} \\ \frac{V}{f'} &= \frac{V + V_s}{f} \\ \frac{f'}{V} &= \frac{f}{V + V_s} \\ f' &= \left\{ \frac{V}{V + V_s} \right\} f\end{aligned}$$

Conclusion

This expression indicates that the frequency of sound as heard by the listener is decreased and he will detect a lower pitch of sound.

CASE 3 : When Both Source And The Listener Are Moving**(A) When Both Source And The Listener Are Moving Towards Each Other**

Let,

f' = Apparent frequency

f = Actual frequency

$V + V_o$ = Speed of sound w.r.t listener

$V + V_s$ = Speed of sound w.r.t source

V = Speed of sound

V_s = Speed of source

V_o = Speed of listener

Physics XI

Then, the apparent frequency heard by the listener when both source and listener are moving towards each other

$$f' = \left\{ \frac{V + V_o}{V - V_s} \right\} f$$

(B) When Both Source And The Listener Are Moving Away From Each Other

Let,

f' = Apparent frequency

f = Actual frequency

$V + V_o$ = Speed of sound w.r.t listener

$V + V_s$ = Speed of sound w.r.t source

V = Speed of sound

V_s = Speed of source

V_o = Speed of listener

Then, the apparent frequency heard by the listener when both source and listener are moving away from each other

$$f' = \left\{ \frac{V - V_o}{V + V_s} \right\} f$$

SHOCK WAVES:

Shock waves are high-energy waves characterized by a sudden and intense increase in pressure, temperature, and density as they travel through a medium. These waves can be generated by a variety of processes and have significant applications in various fields, including physics, engineering, and medicine. Here are some key aspects of shock waves:

Formation: Shock waves typically form when an object or source moves through a fluid (such as air or water) at a speed faster than the speed of sound in that medium. This creates a pressure front that moves outward from the source.

Characteristics: Shock waves are characterized by a steep pressure rise, a rapid increase in temperature, and a sharp density change across the shock front.

Applications:

Aerospace: Shock waves are encountered in supersonic and hypersonic flight, leading to challenges in designing aircraft and spacecraft that can withstand the associated high temperatures and pressures.

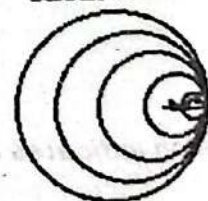
Medicine: Shock waves are used in lithotripsy, a medical procedure to break down kidney stones and other calcified deposits in the body.

Physics: Shock tubes and shock tunnels are used in experimental physics to study the behavior of gases under extreme conditions.

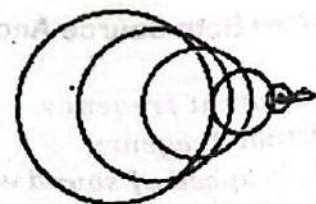
Military: Shock waves from explosions can cause damage and injury, making their understanding important for defense applications.

Sonic Boom: When an aircraft travels at supersonic speeds, it generates shock waves that unite into a single, concentrated shock front called a "sonic boom."

Shock Waves



Aircraft moving at the speed of sound



Aircraft moving faster than the speed of sound

APPLICATION OF DOPPLER'S EFFECT:

ECHOLOCATION:

A physiological process by which an animal emits sound waves and then interprets the echoes to determine the location, size, shape, and movement of objects is called echolocation. Echolocation is used for orientation, obstacle avoidance, food procurement, and social interactions.

CH#12: Acoustics

ASTRONOMY:

Doppler effect is used to measure the motion of stars, galaxies, and other astronomical objects. When a star is moving towards us, the light waves emitted by the star are compressed, resulting in a higher frequency (blueshift). When a star is moving away from us, the light waves are stretched, resulting in a lower frequency (redshift). The amount of redshift or blueshift can be used to determine the speed of the object. For example, if a star is moving away from us at a speed of 100 kilometers per second, the light from the star will be redshifted by a small amount.

SONAR:

When a sound wave is emitted from a sonar device, it travels through the water and bounces off objects. The sound waves that are reflected back to the sonar device are called echoes. The time it takes for the echoes to return tells the sonar device how far away the object is and velocity can be determined by observing changing frequency of echoes.

VOR:

The Doppler effect is used in VOR (VHF Omnidirectional Range) navigation systems to measure the direction of the aircraft relative to the VOR station. A VOR station transmits a signal that consists of two components: a reference signal and a variable signal. The reference signal is a constant frequency, while the variable signal is modulated with a sine wave. The frequency of the variable signal is determined by the direction of the aircraft relative to the VOR station.

RADARS:

The Doppler effect is used in radar to measure the speed and direction of objects. Radar is a system that uses radio waves to detect and track objects.

When a radar transmitter emits a radio wave, it travels through the air and bounces off objects. The radio waves that are reflected back to the radar transmitter are called echoes. The time it takes for the echoes to return tells the radar transmitter how far away the object is.

The Doppler effect can be used to measure the speed of an object by comparing the frequency of the emitted radio waves to the frequency of the reflected radio waves. When an object is moving towards the radar transmitter, the reflected radio waves have a higher frequency than the emitted radio waves. This is because the radio waves are compressed as they travel towards the object.

When an object is moving away from the radar transmitter, the reflected radio waves have a lower frequency than the emitted radio waves. This is because the radio waves are stretched as they travel away from the object.

The amount of change in frequency can be used to determine the speed of the object. For example, if the frequency of the reflected radio waves is 10% higher than the frequency of the emitted radio waves, then the object is moving towards the radar transmitter at a speed of 10% of the speed of light.

The Doppler effect is a powerful tool that can be used to measure the speed and direction of objects. It is used in a variety of applications, such as weather forecasting, traffic control, and military applications.

Radar can be used for:

- Measuring the speed of aircraft:
- Measuring the speed of vehicles
- Detecting weather patterns:
- Tracking the movement of objects:

DOPPLER'S ULTRASOUND:

Doppler ultrasound, often referred to as Doppler ultrasound imaging or simply Doppler imaging, is a medical imaging technique that uses the Doppler effect to assess the blood flow in blood vessels and other structures within the body. The Doppler effect, in this context, provides information about the direction and speed of blood flow.

Chapter 12

Multiple Choice Questions

MCQ'S		ANS.
1. The speed v of a wave represented by $y = A \sin(\omega t - kx)$ is:	a) k/ω b) ω/k c) ωk d) $1/\omega k$	b
2. Two sound waves are $y = A \sin(\omega t - kx)$ and $y = A \cos(\omega t - kx)$. The phase difference between the two waves is:	a) $\pi/2$ b) $\pi/4$ c) π d) 0	a
3. If v_a , v_h and v_m are the speeds of sound in air, hydrogen gas, and a metal at the same temperature, then:	a) $v_a > v_h > v_m$ b) $v_m > v_h > v_a$ c) $v_h > v_m > v_a$ d) $v_h > v_a > v_m$	b
4. The speed of sound in air at STP is 332 m/s. If the air pressure becomes double at the same temperature, the speed of sound become:	a) 1382 m/s b) 664 m/s c) 332 m/s d) 166 m/s	c
5. How does the speed of sound v in air depend on the atmospheric pressure P :	a) $v \propto P^0$ b) $v \propto P^{-1}$ c) $v \propto P^2$ d) $v \propto P^1$	a

6. The speed of sound in a gas is proportional to: a) square root of isothermal elasticity b) square root of adiabatic elasticity c) isothermal elasticity d) adiabatic elasticity	b
7. The length of a pipe closed at one end is L . In the standing wave whose frequency is 7 times the fundamental frequency, what is the closest distance between nodes? a) $\frac{1}{14} L$ b) $\frac{1}{7} L$ c) $\frac{2}{7} L$ d) $\frac{4}{7} L$	c
8. A 620 Hz frequency song of a ice cream trolley approaches with speed v to a boy standing at the door of his house is heard with frequency f_1 . If the trolley is stopped and the boy approaches to the ice cream trolley with same speed v ; the boy now hears the sound with frequency f_2 . choose the correct statement: a) $f_1 = f_2$; both are greater than 620 Hz b) $f_2 > f_1 > 620$ Hz c) $f_1 = f_2$; both are lesser than 620 Hz d) $f_1 > f_2 > 620$ Hz.	a
9. The speed of sound in a gas in which two waves of wavelength 50 cm and 50.4 cm produce 6 beats per second is: a) 338 m/s b) 350 m/s c) 378 m/s d) 400 m/s	c
10. The speed of a wave in a medium is 760 m/s. If 3600 waves are passing through a point in the medium in 2 minutes, then its wave length is: a) 13.85 m b) 25.3 m c) 41.5 m d) 57.2 m	b